Part I

For fill-in-the-blank questions, each blank is worth a point. For true-or-false questions, write only T or **F**. For identification questions, each item is worth a point.

1. The **Free-Body Diagram (FBD)** of a rigid body represents it as being isolated from its surroundings, replacing physical supports with forces and couple moments.
2. *True or False.* An ideal truss does not have three-force members. **T**
3. For 3D systems, the internal moment has two components: **Bending** and **Torsional**, whose axes are tangent and normal to the cross-section, respectively.
4. A body is said to be **Under** constrained if it experiences fewer reactive forces than applicable equations of static equilibrium.
5. *True or False.* A beam supported by a pin (2), a rocker (1), and a roller (1) is statically determinate. **F**
6. For a simply supported 13-member plane truss to be considered simple, it must have **8** joints.
7. *True or False.* A frictionless ball-and-socket resists both rotation and translation. **F**
8. Body where all reactive forces intersect at one point or pass through a common axis is **Improperly** constrained.
9. *True or False.* A stable body requires that the lines of action of the reactive forces do not intersect a common axis and are not parallel to one another. **T**
10. For internal loadings of 2D systems, the **Normal** force and the **Shear** force are tangent and normal to the cross-section, respectively.

Scenario 1

A 1.1-m long cantilever beam is subject to a 2-kN weight at its free end.

11. *True or False.* This scenario is statically determinate.**T**
12. The support reactions have magnitudes of **2** kN (vertical reaction at the fixed support) and **2.2** kN·m (reaction moment at the fixed support).
13. The maximum bending moment developed is **2.2** kN·m in magnitude and occurs **1.1** m from the free end (at the fixed support).
14. The maximum shear force developed is **2** kN in magnitude and occurs **0–1.1** m from the free end (the shear is constant along the cantilever).
15. The graph of the shear force as a function of location along the length of the beam is that of a polynomial of degree **0**.
16. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree **1**.

Scenario 2

A 1-m long beam carries a load uniformly distributed throughout its length (1 kN/m). Assume a roller support 0.1 m from the left end and a pin support at the right end.

17. The roller support develops a force of **0.556** kN.
18. The pin support develops a force of **0.444** kN.
19. The bending moment developed at the right end is **0** kN·m.
20. The shear force developed just to the left of the pin is **0.444** kN.
21. The shear force developed just to the right of the pin is **0** kN.
22. *True or False.* There is zero shear force developed at a point 0.9 m from the right end.**F**
23. The graph of the shear force as a function of location along the length of the beam is that of a polynomial of degree **1**.
24. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree **2**.

Part II:

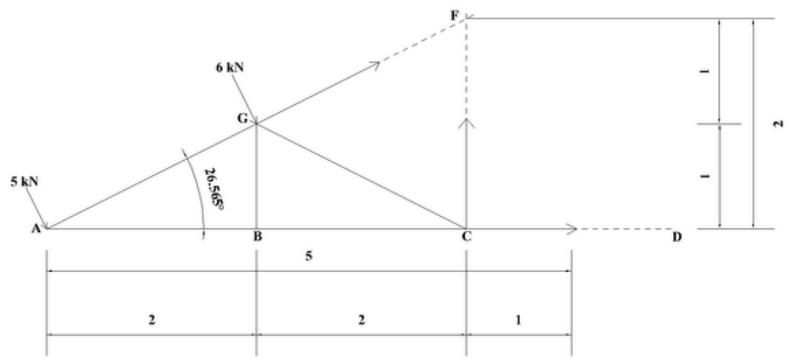
Instructions: Where applicable, provide force diagrams and indicate the sense of member forces. Enclose final answers in boxes, with non-integer values rounded to three (3) decimal places.

Problem 1: Truss Analysis

Given Parameters:

- **Loads:** $p_1 = 5$ kN, $p_2 = 6$ kN, $p_3 = 7$ kN.
- **Geometry:** Panel width $d = 2$ ft, Truss height $h = 3$ ft.
- **Slope Properties:** Horizontal run $= 4$ ft, Rise $= 2$ ft. Hypotenuse $= \sqrt{5}$.

$$\cos\theta \approx 0.8944, \quad \sin\theta \approx 0.4472$$



Solution and Justification

Decomposition of Loads

The loads act perpendicular to the top chord. Using the slope geometry ($\hat{n} = \sin\theta\hat{i} - \cos\theta\hat{j}$):

- $\vec{p}_1 = 2.236\hat{i} - 4.472\hat{j}$ kN
- $\vec{p}_2 = 2.683\hat{i} - 5.367\hat{j}$ kN
- $\vec{p}_3 = 3.130\hat{i} - 6.261\hat{j}$ kN

Method of Sections (Part a)

We cut members **FG**, **CF**, and **CD** and analyze the **Left Section**.

A. Find F_{FG} (Sum Moments about C):

$$\sum M_C = M_{p1} + M_{p2} + M_{FG} = 0$$

$$17.888 + 8.051 - 1.789F_{FG} = 0 \Rightarrow \boxed{F_{FG} = 14.500 \text{ kN (T)}}$$

B. Find F_{CD} (Sum Moments about F):

$$\sum M_F = M_{p1} + M_{p2} - 2F_{CD} = 0$$

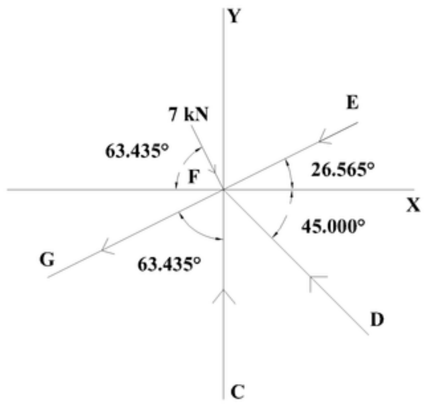
$$22.360 + 13.417 - 2F_{CD} = 0 \Rightarrow \boxed{F_{CD} = 17.889 \text{ kN (C)}}$$

C. Find F_{CF} (Sum Vertical Forces):

$$\sum F_y = \sum p_y + F_{FG}\sin\theta + F_{CF} = 0$$

$$-9.839 + 14.5(0.447) + F_{CF} = 0 \Rightarrow \boxed{F_{CF} = 3.354 \text{ kN (T)}}$$

Method of Joints (Part b)
Isolate Joint F. Knowns: F_{FG} (14.5 T), F_{CF} (3.35 T). Load: p_3.



$$\sum F_x = 3.130 - 12.97 + 0.894F_{EF} + 0.707F_{DF} = 0$$
$$\sum F_y = -6.261 - 6.48 - 3.354 + 0.447F_{EF} - 0.707F_{DF} = 0$$

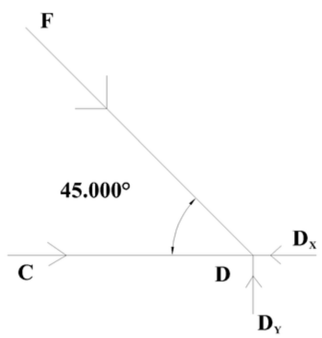
Solving simultaneous equations yields:

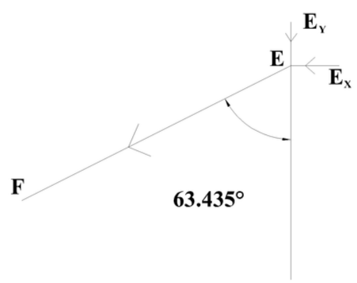
$$\boxed{F_{EF} = 19.330 \text{ kN (T)}} \quad \text{and} \quad \boxed{F_{DF} = 10.520 \text{ kN (C)}}$$

Stability Contingencies (Parts c & d)
Stability: Place a Pin at D and a Roller at E (oriented to provide horizontal reaction). *Justification:* A pin at D resists translation (x, y). A horizontal roller at E creates a force couple with D to resist the overturning moment caused by the cantilever loads.
Zero-Force Members ($p_1 = 0$)**:** If Joint A is unloaded, members AB and AG must be zero. Consequently, members BC and BG at Joint B become zero. **Answer: AB, AG, BG, BC.**

Support Reactions (Part e)
1. Reaction at E:
The support balances F_{EF} (19.330 kN, Tension) which acts at angle θ.
- **Horizontal** (R_{Ex})**:** $F_{EF}\cos\theta = 19.330(0.894) = \boxed{17.281 \text{ kN}}$
- **Vertical** (R_{Ey})**:** $F_{EF}\sin\theta = 19.330(0.447) = \boxed{8.640 \text{ kN}}$
- **Resultant** (R_E)**:** $F_{EF} = \boxed{19.330 \text{ kN}}$

2. Reaction at D:
The reaction balances components of F_{CD} and F_{DF}.
- **Horizontal** (R_{Dx})**:** $17.889 - 10.520\cos(45°) = 17.889 - 7.439 = \boxed{10.450 \text{ kN}}$
- **Vertical** (R_{Dy})**:** $10.520\sin(45°) = \boxed{7.440 \text{ kN}}$
- **Resultant** (R_D)**:** $\sqrt{(10.450)^2 + (7.440)^2} = \sqrt{164.557} = \boxed{12.830 \text{ kN}}$





Problem 2: Beam Analysis

Problem Setup and Coordinate System

We establish a global coordinate system where x represents the distance from the left end of the beam. Using the provided parameters:

- **Start of Beam:** $x = 0$ ft.
- **Bracket Attachment Point (B):** Located at $d_1 + d_2 = 3.2 + 0.8 = \mathbf{4.0}$ **ft.**
- **Support S_1 (Pin):** Located at $x_B + d_3 = 4.0 + 2.0 = \mathbf{6.0}$ **ft.**
- **End of Distributed Load (D):** Located at $x_{S1} + d_4 = 6.0 + 6.0 = \mathbf{12.0}$ **ft.**
- **Support S_2 (Roller):** Located at $x_D + d_5 = 12.0 + 2.0 = \mathbf{14.0}$ **ft.**

Equivalent Loading Calculation

The concentrated load of 5 kips is applied via a bracket at $x = 4.0$. We must resolve this into forces and moments acting *on the beam axis*.

Force Decomposition: The load slope is 3 Vertical : 4 Horizontal.

$$F_x = 5 \times \frac{4}{5} = 4 \text{ kips} \quad \text{(Pointing Left)}$$

$$F_y = 5 \times \frac{3}{5} = 3 \text{ kips} \quad \text{(Pointing Down)}$$

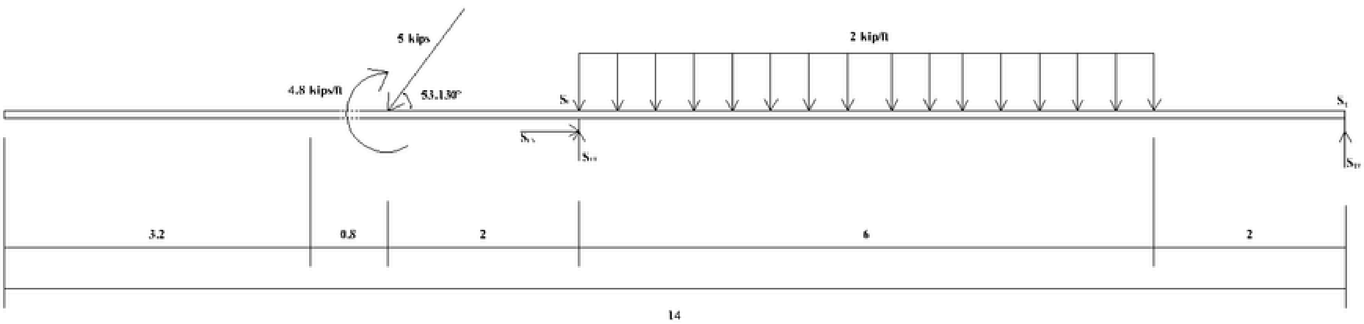
Moment Calculation: We move these forces from the bracket tip to the beam axis ($x = 4.0$).

- Vertical component lever arm (horizontal distance): $d_2 = 0.8$ ft.
- Horizontal component lever arm (vertical height): $d_6 = 0.6$ ft.

Both components create a Counter-Clockwise (CCW) rotation about the attachment point:

$$M_{applied} = (3 \text{ kips})(0.8 \text{ ft}) + (4 \text{ kips})(0.6 \text{ ft})$$

$$M_{applied} = 2.4 + 2.4 = \mathbf{4.8} \text{ k-ft (CCW)}$$



a. Support Reactions

We treat the beam as a rigid body to find external reactions at S_1 (Pin at $x = 6$) and S_2 (Roller at $x = 14$).

Sum of Moments about S_1 ($x = 6$): Taking CCW as positive:

- Moment from Point Load ($x = 4$): Force 3 kips, arm $(6 - 4) = 2$ ft. (Rotation is CCW) $\rightarrow +(3)(2) = +6$.
- Concentrated Couple ($x = 4$): $+4.8$ (Pure moment).
- Moment from Dist. Load ($x = 6$ to 12): Total load $W = 2(6) = 12$ kips. Acts at centroid $x = 9$ (arm 3 ft). (Rotation is CW) $\rightarrow -(12)(3) = -36$.
- Moment from Reaction S_{2y} ($x = 14$): Arm $(14 - 6) = 8$ ft. (Rotation is CCW) $\rightarrow +S_{2y}(8)$.

$$\sum M_{S1} = 6 + 4.8 - 36 + 8S_{2y} = 0$$

$$-25.2 + 8S_{2y} = 0 \implies S_{2y} = \frac{25.2}{8} = \boxed{3.15 \text{ kips}}$$

Sum of Vertical Forces:

$$\sum F_y = S_{1y} + S_{2y} - F_y - W_{dist} = 0$$

$$S_{1y} + 3.15 - 3 - 12 = 0$$

$$S_{1y} + 3.15 - 15 = 0 \implies S_{1y} = 15 - 3.15 = \boxed{11.85 \text{ kips}}$$

Sum of Horizontal Forces:

$$\sum F_x = S_{1x} - F_x = 0 \implies S_{1x} - 4 = 0 \implies \boxed{S_{1x} = 4.00 \text{ kips}}$$

b. Normal Force at $d_1 + d_2$ ($x = 4.0$ ft)

The horizontal component of the load (4 kips) acts at $x = 4$. This force is resisted by the pin support at $x = 6$. Therefore, the beam segment between $x = 4$ and $x = 6$ is in tension.

$$N = \boxed{4.000 \text{ kips (Tension)}}$$

c. Shear Force at $d_1 + d_2$ ($x = 4.0$ ft)

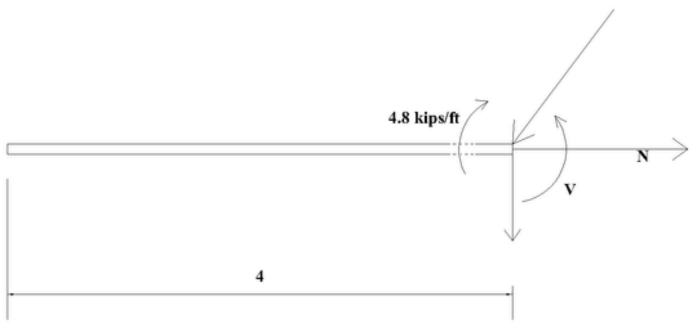
The vertical component of the load (3 kips) acts downward at $x = 4$. Just to the right of this point, the internal shear must balance this load.

$$V = \boxed{-3.000 \text{ kips}}$$

d. Bending Moment at $d_1 + d_2$ ($x = 4.0$ ft)

The bracket creates a concentrated moment of 4.8 k-ft (Counter-Clockwise) at $x = 4$. In beam sign convention, an external CCW couple causes a drop (negative jump) in the moment diagram.

$$M = \boxed{-4.800 \text{ k-ft}}$$



Shear and Moment Functions

Region 1: $0 \le x < 4$ (Free End to Load)

No external loads exist. $V(x) = 0$ and $M(x) = 0$.

Region 2: $4 < x < 6$ (Load to Support S_1)

The vertical load (3 kips) drops shear to $V(x) = -3$. Integrating yields $M(x) = -3x + C_1$.
Boundary Condition: Concentrated couple (4.8 k-ft) at $x = 4$ causes a drop: $M(4) = -4.8$.

$$-3(4) + C_1 = -4.8 \implies C_1 = 7.2 \quad \therefore \quad M(x) = \mathbf{-3x + 7.2}$$

Region 3: $6 < x < 12$ (Support S_1 to End of Dist. Load)

Shear jumps by reaction $S_{1y} = 11.85$: $V(6) = -3 + 11.85 = 8.85$. Load (2 k/ft) applies slope:

$$V(x) = 8.85 - 2(x - 6) = \mathbf{20.85 - 2x}$$

Integrating shear yields $M(x) = 20.85x - x^2 + C_2$. Continuity at $x = 6$ ($M(6) = -10.8$) gives:

$$20.85(6) - 6^2 + C_2 = -10.8 \implies C_2 = -99.9 \quad \therefore \quad M(x) = \mathbf{-x^2 + 20.85x - 99.9}$$

Region 4: $12 < x \leq 14$ **(End of Dist. Load to Roller)**
Shear is constant at end of load: $V(12) = \mathbf{-3.15}$. Integrating yields $M(x) = -3.15x + C_3$.
Continuity at $x = 12$ ($M(12) = 6.3$) gives:

$$-3.15(12) + C_3 = 6.3 \implies C_3 = 44.1 \quad \therefore \quad M(x) = \mathbf{-3.15x + 44.1}$$

Summary of Piecewise Functions

e. Shear Force $V(x)$ **(kips):**

$$\boxed{V(x) = \begin{cases} 0 & 0 \leq x < 4 \\ -3 & 4 < x < 6 \\ 20.85 - 2x & 6 < x < 12 \\ -3.15 & 12 < x \leq 14 \end{cases}}$$

f. Bending Moment $M(x)$ **(k-ft):**

$$\boxed{M(x) = \begin{cases} 0 & 0 \leq x < 4 \\ -3x + 7.2 & 4 < x < 6 \\ -x^2 + 20.85x - 99.9 & 6 < x < 12 \\ -3.15x + 44.1 & 12 < x \leq 14 \end{cases}}$$

Analysis of Key Points

Zero Shear Location: Occurs in Region 3 where $V(x) = 0$.

$$20.85 - 2x = 0 \implies 2x = 20.85 \implies x = \boxed{10.425 \text{ ft}}$$

Maximum Bending Moment: The maximum moment occurs where shear is zero ($x = 10.425$).

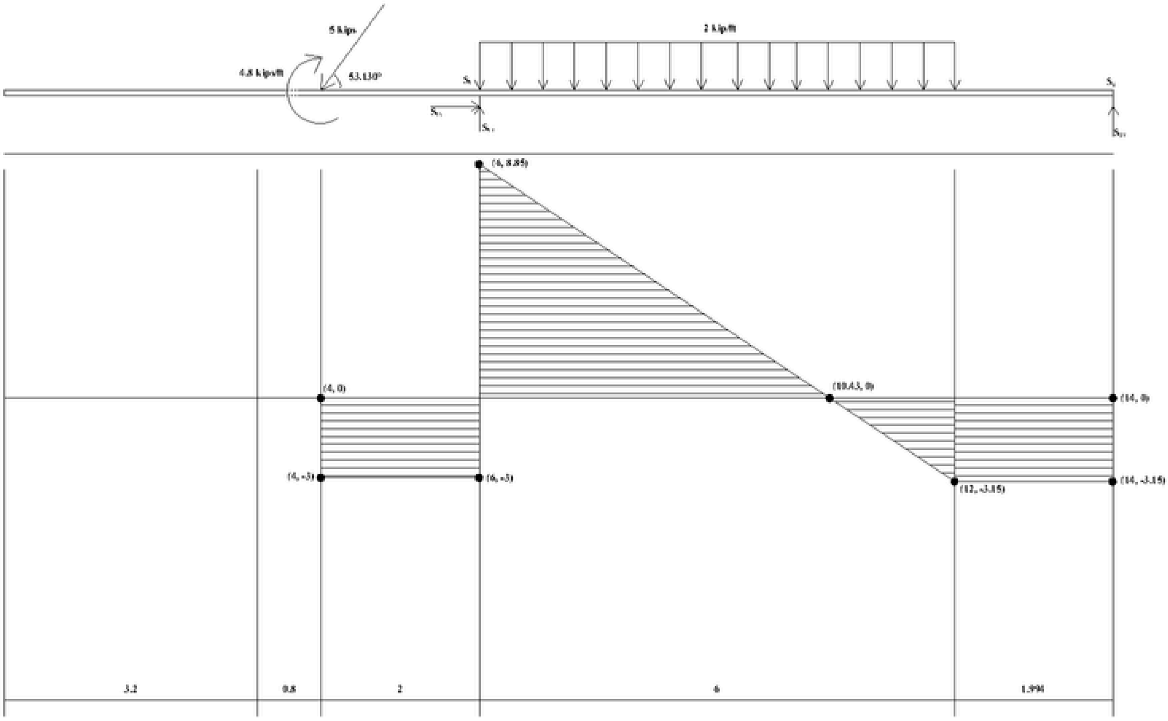
$$M_{max} = -(10.425)^2 + 20.85(10.425) - 99.9$$

$$M_{max} = -108.68 + 217.36 - 99.9 = \boxed{8.78 \text{ k-ft}}$$

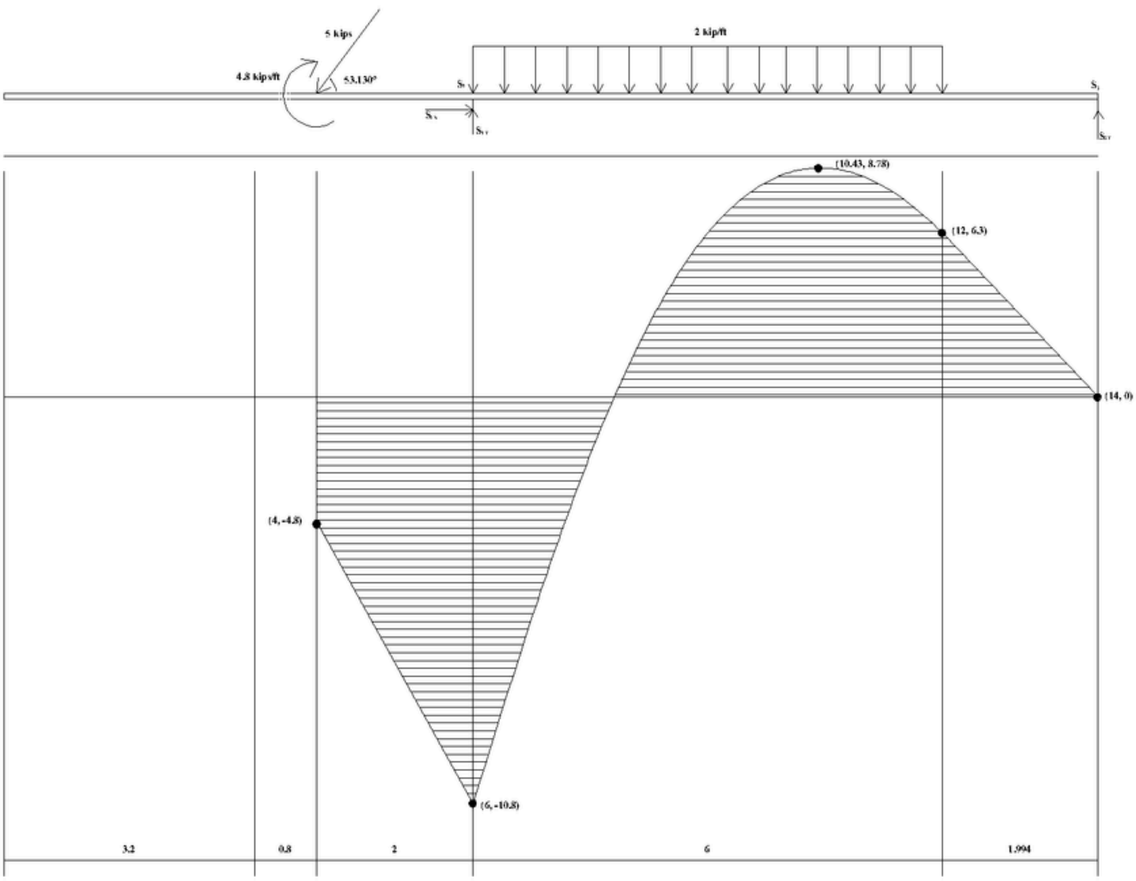
Closure Check at End ($x = 14$)**:**

$$M(14) = -3.15(14) + 44.1 = -44.1 + 44.1 = 0$$

Shear and Moment Diagrams



Shear Force Diagram



Bending Moment Diagram